Spot Challenge - Marker Test Guide (Windows)

This guide shows you how to set up the camera and test ArUco markers. **You do not need to write or change any code.** You only copy commands, paste them into the Command Prompt and press Enter.

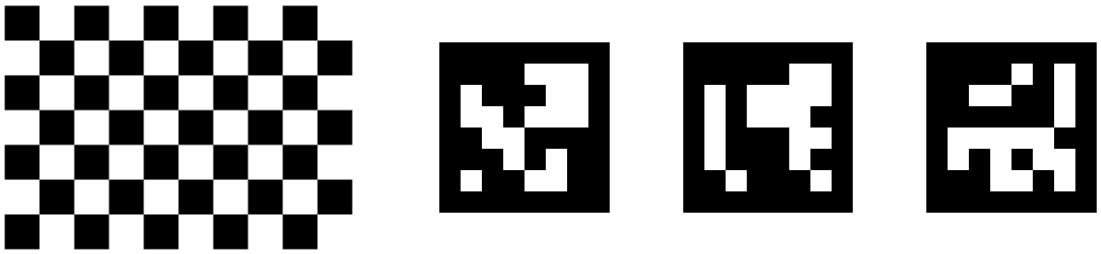
Part A uses the laptop webcam (practice). **Part B** uses the Tello's downward camera. **The drone never takes off in this guide.** The motors stay off.

What you need

1. Internet and a web browser (to download the kit)
2. A printer (A4 paper), a ruler, a measuring tape
3. Flat cardboard and glue or tape
4. A laptop with a webcam and internet (for step 2)
5. A charged Tello drone (the downward camera needs **Tello EDU / Tello TT**)

Important values (keep this page)

Item	Value	File to print
Chessboard	10 x 7 squares (9 x 6 inner corners) square size = 21 mm	print/chessboard_10x7_21mm_A4.pdf
ArUco markers	Dictionary (family): DICT_6X6_250 IDs: 0, 1, 2 black square = 150 mm	print/aruco_DICT_6X6_250_id0-2_150mm_A4.pdf



How to run a command

1. Copy one grey command box, paste it into the Command Prompt, press **Enter**.
2. A program opens a window. To close it: click on the window and press **Q**.
3. If a command shows red error text, look at "Problems and fixes" at the end.

Step 0 - Download the kit

1. Open this link in a web browser: **https://github.com/chissanupong-s/Spot_challenge_fidutial_marker**
2. Click the green **"<> Code"** button, then click **"Download ZIP"**.
3. Open your **Downloads** folder. Right-click `Spot_challenge_fidutial_marker-main.zip` and choose **"Extract All..."**, then click **"Extract"**.
4. Move the unzipped folder `Spot_challenge_fidutial_marker-main` to your Desktop (easy to find).
5. Windows often puts a folder inside a folder with the same name. Open folders until you see the files `GUIDE_Windows.pdf` and `requirements.txt` - that is the kit folder.
6. Open it. You should see the guides, a `print` folder and a `scripts` folder.

Important: Always work inside the unzipped folder, not inside the .zip file.

Step 1 - Print

1. Open `print/chessboard_10x7_21mm_A4.pdf` and `print/aruco_DICT_6X6_250_id0-2_150mm_A4.pdf`.
2. In the print window choose **"Actual size"** or **"100%"**. Do NOT choose "Fit to page".
3. **Measure with a ruler:** one chessboard square should be **21 mm**. The black ArUco square should be **150 mm**.
4. **Write down what you measured.** If it is a bit different (for example 20.5 mm), that is OK - you will type your measured number later.
5. Glue the chessboard **flat** on cardboard. It must not bend.
6. Cut the markers on the dashed line. **Keep the white border** around the black square. Do not fold them.

Note: Meters = millimeters / 1000. Example: 21 mm = 0.021, 20.5 mm = 0.0205, 150 mm = 0.15.

Step 2 - Install (only once)

Important: Do this while the laptop is connected to the normal internet (not the Tello Wi-Fi).

1. Download Python 3.12 from **python.org/downloads** and run the installer.
2. On the first screen **tick "Add python.exe to PATH"**, then click "Install Now".
3. Open the folder `Spot_challenge_fidutial_marker-main` (from Step 0).
4. Click the address bar at the top of the folder window, type `cmd` and press Enter. A black window (Command Prompt) opens in this folder. Use Command Prompt, not PowerShell.
5. Paste these commands one by one:

```
python -m venv .venv
.venv\Scripts\activate
python -m pip install --upgrade pip
pip install -r requirements.txt
```

When you see **(.venv)** at the start of the line, you are ready.

Important: Every time you open a new Command Prompt: open it in the kit folder (as above) and run this first:

```
.venv\Scripts\activate
```

Part A - Laptop webcam

Step 3 - Find the camera ID

A laptop can have more than one camera. Each camera has a number (ID): 0, 1, 2 ...

```
python -m scripts.aruco.check_cameras
```

1. A window shows the camera image with **CAMERA ID = 0**.
2. Press **N** to see the next camera. Press **Q** to close.
3. **Write down the ID** of the camera you want (usually 0 = built-in, 1 = USB camera).
4. In the next commands, replace `--camera-id 0` with your ID.

Step 4 - Take chessboard photos

```
python -m scripts.aruco.capture_calibration_images --camera webcam --camera-id
0
```

1. Hold the chessboard in front of the camera.
2. Green text **"BOARD FOUND"** = good: hold still and press **SPACE** to save a photo.
3. Red text = the camera cannot see the whole board. Move it until the text is green.
4. Take **20 to 30 photos**: near and far, tilted left / right / up / down, and in every corner of the picture.
5. Use good light. Avoid shiny reflections on the board.
6. Press **Q** when done.

Step 5 - Calibrate the webcam

```
python -m scripts.aruco.calibrate_camera --camera webcam --square-size 0.021
```

1. `--square-size` = **your measured square size in meters** (21 mm = 0.021).
2. Wait until it prints **"Calibration complete."**
3. Look at the line **"Mean reprojection error (pixels)"**: below 0.5 = great, below 1.0 = OK, above 1.0 = do Step 4 again (first delete the photos in `data/calib_images/webcam`).
4. The result is saved automatically in `assets/calibration/webcam`. You do not need to copy anything.

Step 6 - Test the printed markers

```
python -m scripts.aruco.webcam_distance --camera-id 0 --marker-length 0.15
```

1. `--marker-length` = **your measured black square in meters** (150 mm = 0.15).
2. Hold marker ID 0 in front of the camera. You see a green outline, colored axes and text like **"ID: 0 | Z: 0.503 m"**. Z = distance from the camera in meters.
3. Check it: put the marker **50 cm** and **100 cm** from the camera (use the tape). Z should be close (within about 2-3 cm). Write the numbers in the results table.
4. Try markers ID 1 and ID 2 too. Press **Q** to close.

Part B - Tello downward camera

Important: The drone does NOT take off in these steps. Hold it in your hand or put it on a box. Use a fully charged battery.

Step 7 - Connect and first test

1. Turn on the Tello and wait about 10 seconds.
2. Connect the laptop Wi-Fi to **TELLO-XXXXXX**. (No internet now - that is normal.)
3. If Windows asks **"Allow access"** (Firewall) for Python: tick **both** boxes (Private and Public) and click Allow. Without this you get no video.
4. Open the Command Prompt in the kit folder, run `.venv\Scripts\activate`, then:

```
python -m scripts.tello.marker_test --camera down
```

1. A small grey picture appears - this is the camera under the drone.
2. Put a marker on the floor. Hold the drone 30-50 cm above it, camera facing down.
3. You see a green outline and text like **"ID:0 area:5200px"**.
4. "Calibration: NO" is normal now - distance comes after Step 9.
5. Press **V** to switch to the front camera and back. Press **Q** to close.

Important: If you see "Tello rejected the camera switch": this drone cannot use the downward camera (it needs Tello EDU / TT). Stop here and tell the dev team.

Step 8 - Chessboard photos with the downward camera

```
python -m scripts.aruco.capture_calibration_images --camera tello_down
```

1. Put the chessboard flat on the floor or a table, in good light.
2. Hold the drone above it, camera facing down, about 30-60 cm high.
3. This time **move and tilt the drone**, not the board.
4. Same rules as Step 4: green "BOARD FOUND" -> press **SPACE**. Take **20 to 30 photos**. Press **Q** when done.

Step 9 - Calibrate the downward camera

```
python -m scripts.aruco.calibrate_camera --camera tello_down --square-size
0.021
```

Check the "Mean reprojection error" the same way as Step 5. The result is saved in `assets/calibration/tello_down`.

Step 10 - Test markers with distance

```
python -m scripts.tello.marker_test --camera down --marker-length 0.15
```

1. Now the screen says **"Calibration: yes"** and shows **z** in meters.

2. Hold the drone **30 cm, 50 cm and 100 cm** above the marker (measure from the camera to the floor). Write down z each time.
3. Press **S** to save a picture at each height.
4. Try markers ID 1 and ID 2, and a marker near the edge of the picture.

Step 11 - Send the results to the dev team

Zip these and send them:

1. the folder `assets/calibration`
2. the folder `data/snapshots`
3. your results table (a photo of the paper is fine)

Camera	Marker ID	Real distance	Screen Z / z	OK?
Webcam	0	50 cm		
Webcam	0	100 cm		
Tello down	0	30 cm		
Tello down	0	50 cm		
Tello down	0	100 cm		

Problems and fixes

Problem	Fix
'python' is not recognized	Install Python again and tick "Add python.exe to PATH". Or type py instead of python.
No module named 'cv2' or 'djitellopy'	You forgot to activate. Run: .venv\Scripts\activate
No module named 'scripts'	The Command Prompt is not in the kit folder. Open it in the folder again.
Camera window is black / does not open	Close Zoom, Teams, browser camera tabs. Try another camera ID.
Text never turns green (Step 4 / 8)	More light, board flat, the WHOLE board in the picture, no reflections, hold still.
"Only N valid calibration images"	Take more photos (Step 4 / 8), then calibrate again.
"Calibration file not found"	Do the calibrate step (Step 5 / 9) first.
Marker is not detected	Show the whole marker with its white border. Keep it flat. Come closer. More light. Only the printed markers from this kit work.
WARNING: calibration ... does not fit	The calibration was made with another camera. Redo the photos + calibrate steps for this camera.
Tello: no picture / stuck	Is the Wi-Fi connected to TELLO-XXXXXX? Turn the drone off and on, wait 10 s, try again. Allow Python in the Windows Firewall.